

PROBLEM STATEMENT	PROPOSED SOLUTION
The Production LLM Overspend & Latency Bottleneck. 70-80% of production LLM queries are simple, conversational or conceptual (e.g. "What is a Python list?"), yet standard architectures forward 100% of traffic to expensive 70B+ cloud endpoints. This causes unnecessary cloud round-trip latency (1.5s-3.5s) for answers that could run locally in milliseconds, wastes completion tokens by discarding failed edge outputs and re-generating from scratch on cloud models, and drains energy by running trivial prompts on 300W+ datacenter GPUs - inflating both cost and carbon footprint.	The TriForge Hybrid Smart Gateway. An intelligent routing gateway sitting between client apps and LLM backends. It inspects query complexity in real time, routing low-risk/simple queries to a free, ultra-fast local edge model (Groq Llama 3.1 8B or local Ollama) while reserving paid remote models (Fireworks AI, GPT-4o, Claude 3.5 Sonnet) strictly for complex code synthesis and advanced reasoning. When a local answer fails verification, the Verify-Draft Escalation Protocol sends the local draft to the cloud verifier to edit rather than regenerate - cutting cloud completion tokens up to 10x.

KEY FEATURES & FUNCTIONALITY	
1. Smart Intent Classification 9 target domains (coding, math, reasoning, summarization, translation, extraction, conversation, creative writing, general QA); conceptual QA routes locally, code synthesis routes remotely.	2. SHA-256 Prompt Caching Pre-inference hash lookup intercepts duplicate requests, returning cached completions in <1ms.
3. Selective Self-Consistency Subjective/conversational queries skip double-sampling (saves 50% tokens); technical QA runs parallel dual-sampling via ThreadPoolExecutor.	4. Verify-Draft Escalation Protocol Local baseline drafts are passed to the cloud verifier for editing instead of full regeneration - cutting remote completion tokens up to 10x.
5. API Key Shield & Privacy Server-side key masking redacts credentials as in frontend responses, preventing client-side leaks.	6. Real-Time Green AI Tracker Dashboard quantifying Energy Conserved (kWh), CO ₂ Emissions Avoided (kg) and phone-charge equivalents offset by edge routing.

PLATFORM & INFRASTRUCTURE	WHAT MAKES IT DIFFERENT
Groq: zero-shot intent classification via Llama 3.1 8B with connection timeouts. Edge Ollama: zero-token on-device inference (Qwen 2.5 / Gemma 2). Pluggable Cloud Fabric: unified middleware for Fireworks AI, OpenAI GPT-4o, Anthropic Claude 3.5, Groq Llama 3.3 70B - no vendor lock-in. Backend: FastAPI, Python 3.11, async event loops, SSE streaming, SQLite (WAL mode, 64MB cache). Frontend: Next.js 14 + TailwindCSS, Recharts telemetry, persistent localStorage sessions.	Verify-Draft Escalation Loop: unlike legacy routers that discard failed local attempts, TriForge sends local output as a draft for the cloud model to edit/correct - saving up to 10x output tokens. Green AI & Sustainability Tracker: first-of-its-kind dashboard quantifying energy conserved, CO ₂ avoided and battery-recharge equivalents from edge offloading. 1-Click Evaluation Exporter: structured Markdown benchmark reports with executive summaries, accuracy matrices, latency breakdowns and eco-impact stats.

ROUTING DECISION STRATEGY MATRIX			
Query Category	Intent / Length Filter	Route	Rationale
Conversation / Greeting	"hello", "hi", etc.	LOCAL EDGE	Bypasses consistency checks; served zero-cost locally
Conceptual Code QA	"What is a Python list?"	LOCAL EDGE	Informational programming QA served locally
Code Synthesis / Debug	"Write a python script..."	REMOTE CLOUD	High-complexity code writing requires 70B+ synthesis
Long Context	> 75 words	REMOTE CLOUD	Prevents local context overflow and latency degradation
Factual QA / Math	Standard prompts	LOCAL → VERIFY	Checks self-consistency; escalates via verify-draft if similarity < 0.8

EXPECTED IMPACT, VALUE & METRICS		
Dimension	Traditional → TriForge	Business Value
API Token Expenditure	\$450/day → \$90/day	\$10,800/mo net savings (80% cut, 100k reqs)
Response Latency	1,500-3,500ms → <1ms cache 320-400ms local	3x-5x latency speedup
Self-Consistency Check	Sequential >1,500ms → <800ms parallel	50% verification speedup
Sustainability Footprint	300W cloud GPU → ~0.0005 kWh saved/request	Active carbon footprint reduction